MUHAMMAD FAHAD PERVEZ

AI & SOFTWARE ENGINEER | BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

Computer Science undergraduate experienced in building AI-powered applications, Retrieval-Augmented Generation (RAG) chatbots, computer vision systems, machine learning solutions, and backend APIs. Skilled in Python, FastAPI, SQL, modern AI frameworks, and cloud technologies, with a focus on production-ready AI systems, backend architecture, and intelligent software solutions.

TECHNICAL SKILLS

Python | SQL (PostgreSQL) | FastAPI | React | Machine Learning | Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | Computer Vision (OpenCV) | LangChain & LangGraph | Docker | AWS (EC2, S3, RDS) | Git & GitHub

PROJECTS AND EXPERIENCE

LookOut AI — AI Surveillance & Intrusion Detection System

Technologies: *Python, FastAPI, OpenCV, PostgreSQL, Docker, AWS*

- Developed a real-time surveillance solution for intrusion detection using computer vision-based person detection and tracking.
- Built FastAPI services to manage security events and stored intrusion records and evidence in PostgreSQL.
- Containerized application services with Docker and integrated AWS cloud services for deployment infrastructure.

ViperBot — Enterprise RAG Chatbot

Technologies: *Python, FastAPI, LangChain, LangGraph, ChromaDB, Gemini*

- Built a document-grounded question-answering chatbot using Retrieval-Augmented Generation (RAG) and semantic search.
- Implemented hybrid retrieval with BM25 and Maximal Marginal Relevance (MMR) reranking to improve context selection.
- Designed LangGraph workflows with conversational memory, generated embeddings for vector search, and exposed capabilities through FastAPI APIs.

RabtaaAI — AI Civic Complaint Platform

Technologies: *React, FastAPI, Supabase, Gemini*

- Developed an AI-assisted civic complaint platform with chatbot support and image understanding capabilities.
- Built React interfaces and connected backend services to Supabase database infrastructure.

Traffic Crash Severity Prediction

Technologies: *Python, Scikit-learn, XGBoost, Pandas*

- Developed machine learning models to predict traffic accident severity using structured accident data.
- Performed data preprocessing, feature engineering, model evaluation, and optimization to improve predictive performance.

EDUCATION

Intermediate in General Sciences

Commecs College | 2021-2023

Bachelor of Science in Computer Science

University of Karachi | 2024 – 2027

CERTIFICATIONS

Python Developer Certification — Udemy (Andrei Neagoie)-2023

Artificial Intelligence & Data Science Program — Saylani Mass IT Training-2026